JavaScript RegExp Methods

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Complete RexExp Reference

The complete RegExp reference includes:

RegExp Methods:

Method	Description
`test()`	Tests for a match - returns true/false
`exec()`	Returns match details or null

String Methods using RegExp:

Method	Description
`match(pattern)`	Returns array of matches
`matchAll(pattern)`	Returns iterator of matches
`replace(pattern, repl)`	Replaces matches
`search(pattern)`	Returns position of match
`split(pattern)`	Splits by pattern

RegExp Object Methods

`test()` and `exec()` are the two RegExp object methods:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Methods</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp Object Methods</h4>
<p id="demo"></p>

<script>
const text = "Hello World";

// test() - returns boolean
const testResult = /World/.test(text);

// exec() - returns match object
const execResult = /World/.exec(text);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "test(/World/): " + testResult + "<br>" +
  "exec(/World/)[0]: " + execResult[0] + "<br>" +
  "exec() index: " + execResult.index;
</script>

</body>
</html>
```

Example 1

Result [View Example]

String Methods with RegExp

String methods that work with RegExp patterns:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Methods</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>String Methods with RegExp</h4>
<p id="demo"></p>

<script>
const text = "JavaScript is awesome, JavaScript!";

let output = "Text: " + text + "<br><br>";
output += "match(/JavaScript/g): " + text.match(/JavaScript/g) + "<br>";
output += "search(/awesome/): " + text.search(/awesome/) + "<br>";

// matchAll returns iterator
const iterator = text.matchAll(/JavaScript/g);
output += "matchAll: ";
for (let m of iterator) output += m[0] + " ";

document.getElementById("demo").innerHTML = output;
</script>

</body>
</html>
```

Example 2

Result [View Example]

String.replace() and split()

`replace()` replaces matches. `split()` splits strings by a pattern:

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Methods</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>String.replace() and split()</h4>
<p id="demo"></p>

<script>
const text = "Hello 123 World";

// replace() with RegExp
const replaced = text.replace(/\d+/, "NUMBERS");

// split() with RegExp
const parts = text.split(/\s+/);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "replace(/\\d+/, 'NUMBERS'): " + replaced + "<br>" +
  "split(/\\s+/): " + parts;
</script>

</body>
</html>
```

Example 3

Result View Example

Pattern-Based Validation

Combine RegExp methods for practical form validation:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Methods</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Pattern-Based Validation</h4>
<p id="demo"></p>

<script>
// Email validation pattern
const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

// Phone validation pattern
const phonePattern = /^\d{3}-\d{3}-\d{4}$/;

let output = "Email: john@example.com → ";
output += emailPattern.test("john@example.com") ? "✓ Valid" : "✗ Invalid";
output += "<br>";

output += "Email: invalid → ";
output += emailPattern.test("invalid") ? "✓ Valid" : "✗ Invalid";
output += "<br><br>";

output += "Phone: 555-123-4567 → ";
output += phonePattern.test("555-123-4567") ? "✓ Valid" : "✗ Invalid";
output += "<br>";

output += "Phone: 5551234567 → ";
output += phonePattern.test("5551234567") ? "✓ Valid" : "✗ Invalid";

document.getElementById("demo").innerHTML = output;
</script>

</body>
</html>
```

Example 4

Result [View Example]

Document

Document in project

You can Download PDF file.

Reference

- W3Schools JavaScript RegExp Methods